

Opinion

Understanding human-commensalism through an ecological and evolutionary framework

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Human-commensalism has been intuitively characterised as an interspecific interaction whereby non-human individuals benefit from tight associations with anthropogenic environments. However, a clear definition of human-commensalism, rooted within an ecological and evolutionary framework, has yet to be proposed. Here, we define human-commensalism as a population-level dependence on anthropogenic resources, associated with genetic differentiation from the ancestral, non-commensal form. Such a definition helps us to understand the origins of human-commensalism and the pace and form of adaptation to anthropogenic niches, and may enable the prediction of future evolution in an increasingly human-modified world. Our discussion encourages greater consideration of the spatial and temporal complexity in anthropogenic niches, promoting a nuanced consideration of human-commensal populations when formulating research questions.

The need to define human-commensalism

Humans have acted as **niche constructors** (see [Glossary](#)) since the Late Pleistocene [1,2]. Through development of fixed settlements, agricultural practices, transportation networks, and energy infrastructure, humans have directly or indirectly modified 95% of the Earth's land [3]. The introduction of anthropogenic selective pressures has had varying impacts on the ecological and evolutionary trajectories of species across different spatial and temporal scales [4–6]. For many species, increased human activity has been associated with population size reductions, extirpations, or extinction [7–10]. For others, trait modification, either via phenotypic **plasticity** or **adaptation**, has enabled persistence in **human-modified environments** [11]. A small number of populations have even benefitted from increased human association, with some evolving a dependency upon **anthropogenic resources** [12].

The term 'human-commensal' has been used to broadly categorise a subset of non-human taxa that benefit from tight human association [13]. Without a rigorous definition, human-commensal status has been somewhat intuitively applied across a variety of taxonomic groups, including amphibians [14], birds [15–19], insects [20–22], primates [23–25], plants [26], and rodents [27–29]. Human-commensalism has also been referenced in multiple ecological contexts, including the study of commensal pathways to speciation and domestication [30–33], dispersal and phylogeographic dynamics of pests [29,34] and disease vectors [27], and phenotypic [35] or genomic [36–38] adaptation to human-modified habitats. Human-commensalism is therefore of ecological and evolutionary interest, but the term's utility is limited by the lack of a clear definition.

Modern understanding of human-commensalism extends beyond the classical cost–benefit frameworks associated with the term 'commensalism' (Box 1). Human-commensalism is now

Highlights

Human-commensalism has been used to broadly describe interactions in which non-human partners benefit from tight associations with humans and their environments.

Human-commensal status has often been intuitively assigned, and the term has been applied without first being clearly defined. This has led to inconsistencies in application across different fields and confusion with other ecologically distinct descriptors of human association.

We propose a clear definition of human-commensalism rooted within an ecological and evolutionary framework. Such framing enables exploration of the potential origins of human-commensalism and has implications for the study of biogeography and demography, the pace and form of adaptation to an anthropogenic niche, and predictions of future evolutionary change in response to anthropogenic pressures.

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